



University Institute of Engineering
Department of Computer Science & Engineering

Experiment-12

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Subject Name: ADBMS

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1. Aim of the practical: To study and implement **PostgreSQL Triggers** using trigger functions for automatically performing actions during INSERT, UPDATE, and DELETE operations, including message generation, validation, and audit logging.

2. Questions

(A) Requirements: DESIGN A PORSTGRESQL TRIGGERS THAT:

Whenever a new employee is inserted in tbl_employee, a record should be added to tbl_employee_audit like:

"Employee name <emp_name> has been added at <current_time>"

Whenever an employee is deleted from tbl_employee, a record should be added to tbl_employee_audit like:

"Employee name <emp_name> has been deleted at <current_time>"

The solution must use PostgreSQL triggers.

```
CREATE OR REPLACE FUNCTION audit_employee_changes()
RETURNS TRIGGER
LANGUAGE plpgsql
AS
$$
BEGIN
    IF TG_OP = 'INSERT' THEN
        INSERT INTO tbl_employee_audit(message)
        VALUES ('Employee name ' || NEW.emp_name || ' has been added at ' || NOW());
        RETURN NEW;

    ELSIF TG_OP = 'DELETE' THEN
        INSERT INTO tbl_employee_audit(message)
        VALUES ('Employee name ' || OLD.emp_name || ' has been deleted at ' || NOW());
        RETURN OLD;
    END IF;

    RETURN NULL;
END;
$$
```



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Solution:

	sno [PK] integer	message text
1	1	Employee name Aman has been added at 2026-09-27 23:10:36.584673+0...

3. Learning Outcomes:

- Learned to create and implement PostgreSQL Triggers.
- automatic actions during INSERT, UPDATE, and DELETE operations.
- Learned to use trigger functions for data validation and processing.
- Applied triggers for maintaining audit records and database integrity.
- Developed practical skills in automating database operations using triggers.